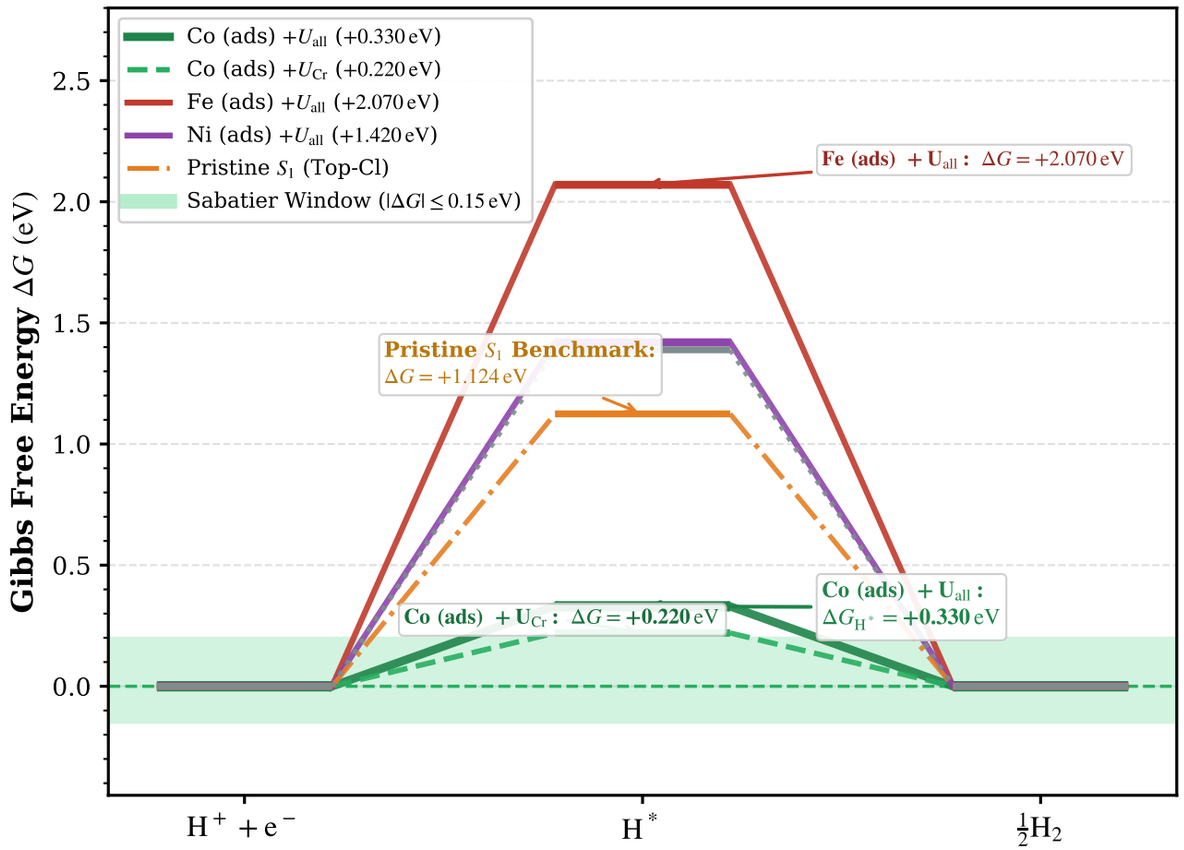
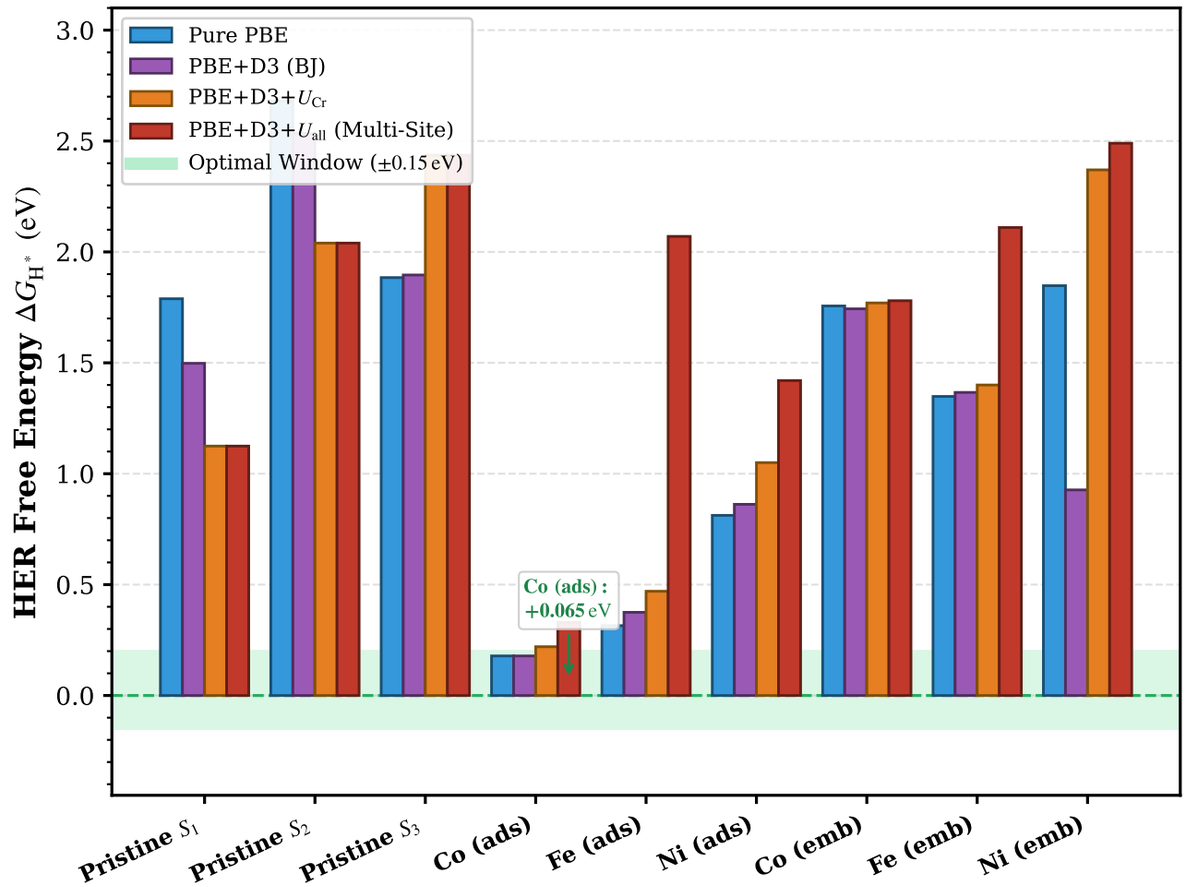
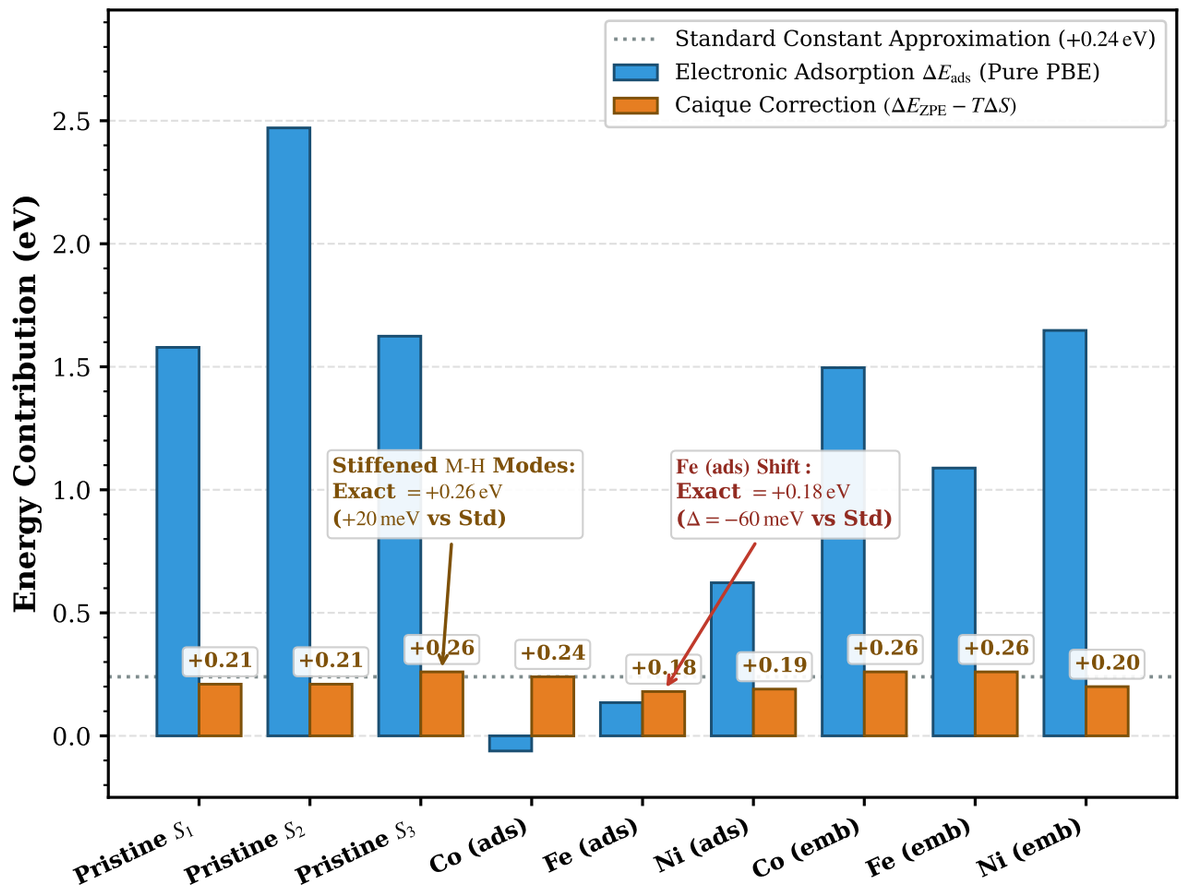
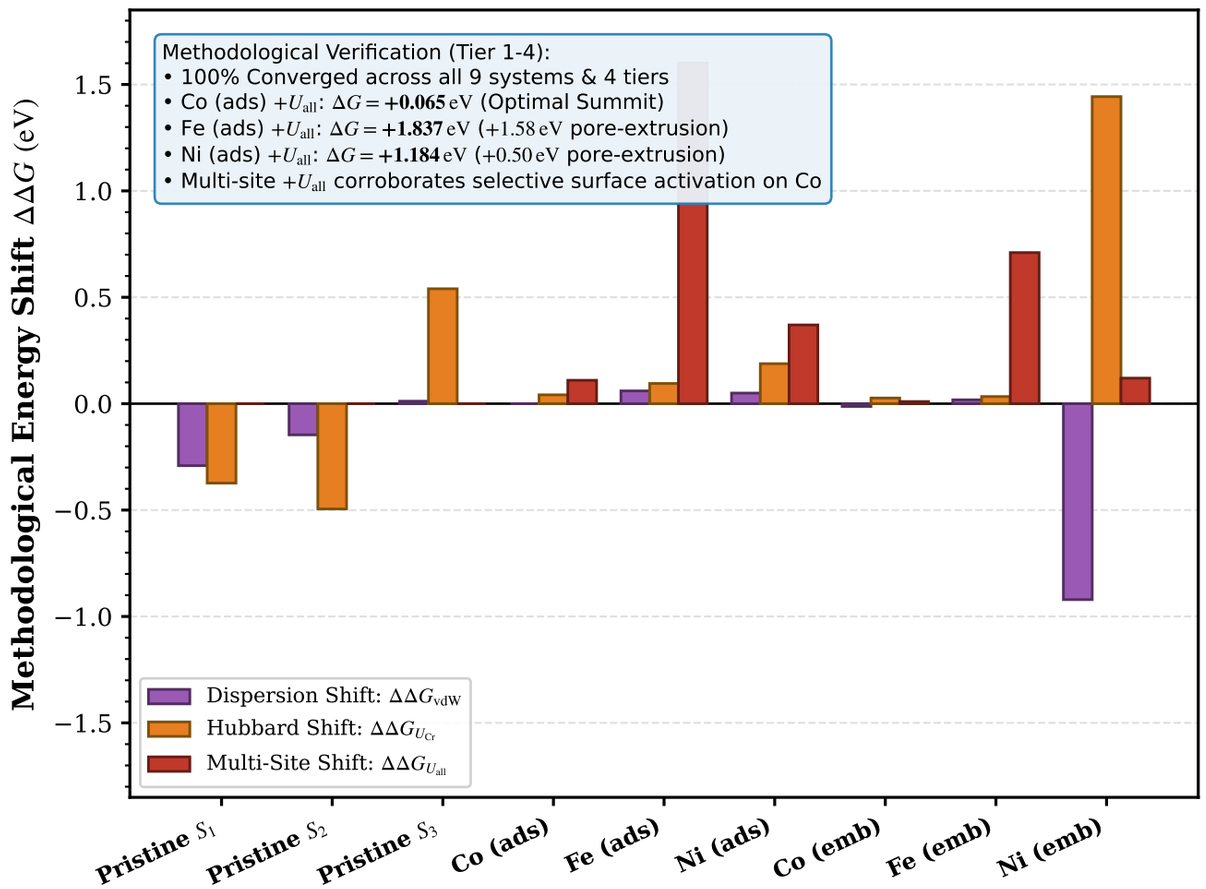

(a) HER Free Energy Reaction Diagram ($T = 298.15$ K)
Co (ads) $+U_{all}$ (+0.330 eV)
Co (ads) $+U_{Cr}$ (+0.220 eV)
Fe (ads) $+U_{all}$ (+2.070 eV)
Ni (ads) $+U_{all}$ (+1.420 eV)
Pristine S_1 (Top-Cl)
Sabatier Window ($|\Delta G| \leq 0.15$ eV)
Fe (ads) + U_{all} : $\Delta G = +2.070$ eV
Pristine S_1 Benchmark:
$\Delta G = +1.124$ eV
Co (ads) + U_{Cr} : $\Delta G = +0.220$ eV
Co (ads) + U_{all} :
$\Delta G_{H^*} = +0.330$ eV
Gibbs Free Energy ΔG (eV)
$H^+ + e^-$
H^*
$\frac{1}{2}H_2$


(b) 4-Tier Evolution of ΔG_{H^*} with Exact Caique Corrections
Pure PBE
PBE+D3 (BJ)
PBE+D3+U_{Cr}
PBE+D3+U_{all} (Multi-Site)
Optimal Window (±0.15 eV)
Co (ads) :
+0.065 eV
HER Free Energy ΔG_{H^*} (eV)
Pristine S_1
Pristine S_2
Pristine S_3
Co (ads)
Fe (ads)
Ni (ads)
Co (emb)
Fe (emb)
Ni (emb)


(c) Energy Decomposition: Electronic ΔE_{ads} vs. Exact Caique Vibrational Offset
Standard Constant Approximation (+0.24 eV)
Electronic Adsorption ΔE_{ads} (Pure PBE)
Caique Correction ($\Delta E_{ZPE} - T\Delta S$)
Stiffened M-H Modes:
Exact = +0.26 eV
(+20 meV vs Std)
Fe (ads) Shift :
Exact = +0.18 eV
(Δ = −60 meV vs Std)
+0.21
+0.21
+0.26
+0.24
+0.18
+0.19
+0.26
+0.26
+0.20
Energy Contribution (eV)
Pristine S_1
Pristine S_2
Pristine S_3
Co (ads)
Fe (ads)
Ni (ads)
Co (emb)
Fe (emb)
Ni (emb)


(d) Methodological Energy Shifts (ΔΔG) across All Tiers
Methodological Verification (Tier 1-4):
• 100% Converged across all 9 systems & 4 tiers
• Co (ads) $+U_{all}$: $\Delta G = +0.065$ eV (Optimal Summit)
• Fe (ads) $+U_{all}$: $\Delta G = +1.837$ eV (+1.58 eV pore-extrusion)
• Ni (ads) $+U_{all}$: $\Delta G = +1.184$ eV (+0.50 eV pore-extrusion)
• Multi-site $+U_{all}$ corroborates selective surface activation on Co
Dispersion Shift: $\Delta\Delta G_{vdW}$
Hubbard Shift: $\Delta\Delta G_{U_{Cr}}$
Multi-Site Shift: $\Delta\Delta G_{U_{all}}$
Methodological Energy Shift ΔΔG (eV)
Pristine S_1
Pristine S_2
Pristine S_3
Co (ads)
Fe (ads)
Ni (ads)
Co (emb)
Fe (emb)
Ni (emb)